

Group 5 Hackathon Report

EDA & Data Preprocessing

For EDA, first we drill down to find the order distribution based on Year, Month, Season or Day of Week to uncover hidden seasonality patterns. We can see that the most orders were placed in 2011, and some months do not have any orders, which is needed to notice.

A major insight emerged while examining **Year-Month distributions**, where several months showed **zero recorded orders**. Specifically, the months **2010-07, 2010-08, 2010-09, 2010-10, 2010-11, and 2011-12** contained *no rows at all* in the dataset. These empty periods are irregular because they break the continuity of the time series and may distort temporal analyses such as seasonality detection, trend modeling, and forecasting. The absence of data could indicate missing records, system outages, or operational pauses rather than true zero-order activity, meaning these gaps must be treated carefully. Their presence implies that models relying on continuous historical patterns may require imputation, exclusion handling, or structural adjustments.

Beyond missing months, additional irregularities were observed in the form of inconsistent review activity, edge-case outliers in lead times and ratings, and occasional gaps in product- or country-level records. Missing values analysis pointed to uneven data completeness across features, suggesting varying reliability of certain fields. These irregularities highlight the importance of data cleaning before modelling, ensuring continuity where needed, detecting bias introduced by gaps, and validating whether the zero-month periods reflect genuine operational conditions or data collection issues. Together, these findings help clarify the dataset's limitations and guide next steps in preparing robust models.

For **data preprocessing**, we performed feature engineering to allow the models to better recognize seasonal, regional and customer behavioural patterns. The following features were developed: temporal features, seasonal indicators, lead time, climate-based groupings, price features, cyclical month encoding, and customer behavioural features. After feature engineering, we applied one-hot encoding to convert categorical variables into a model-ready numerical format.

Uncertainty 1 - Number of Distinct Orders Model

To prepare the dataset for SARIMA and ARIMA, we reconstructed a continuous monthly series by reindexing the data, interpolating missing months, and removing zero-filled gaps. These steps were necessary to make both classical time-series models operational; however, they also underscored key limitations in the underlying data. The short historical window, irregular ordering patterns, and high volatility prevented SARIMA and ARIMA from learning stable trends or seasonality, resulting in very high MAPE values and non-significant parameters. In contrast, XGBoost performed substantially better because it does not rely on strict temporal continuity and can learn non-linear patterns from the available features despite the irregularities. Overall, the results indicate that the weak performance of SARIMA and ARIMA stems from data constraints rather than model specification.

Model	MAPE
SARIMA	1966.67

ARIMA	6266.67
XG BOOST	169.00

Uncertainty 2 - Product Classification Model

For Model 2, our target was the Product Code. We applied and compared a multinomial logistic regression model, Random Forest and XG Boost. In all our models, we accounted for class imbalance. Additionally, for random forest and XG Boost we tuned both models to try and achieve the best performing model using GridSearchCV and RandomSearchCV. The performance of the models were assessed using the accuracy matrix. The results were as follows:

Model	Accuracy
Logistic Regression	21.04%
Random Forest	13.21%
Tuned Random Forest	12.07%
XG Boost	24.80%
Tuned XG Boost	20.07%

We find that the best performing model is the **XG Boost** model. However, the overall accuracy of the models tested were very low and we believe that this is a direct impact of the data quality issues that we highlighted in the EDA portion of our report.

Uncertainty 3 - Number of Items Model

To predict the quantity of items per product our target variables were Items. The models used were Random Forest and XGBoost, these were tuned to achieve the best performing model, by using GridSearchCV and RandomizedSearchCV, respectively. To assess the performance of each model, we computed MAPE, MAE, RMSE and R-squared. We then selected the best performing model based on the MAPE score, the results were as follows:

Model	MAPE
Random Forest	289.92%
Tuned Random Forest	226.99%
XGBoost	229.48%
Tuned XGBoost	240.47%

The results revealed high MAPE scores for both models, and both their tuned versions, with Tuned Random Forest as our best model, achieving a MAPE score of 226.99%.

Uncertainty 4 - Lead Time Prediction Model

Model 4 involved training and testing three models: Linear Regression, Random Forest, and XGBoost. First, we prepared the data before feeding it into each model by assigning the target as lead days, calculated by subtracting the order date from the requested delivery date. Next, we removed any invalid values (0 or negative, if applicable), applied one-hot encoding to categorical variables, and kept only those aligned with the target variables. We split the data similarly to the other models.

We tried the empirical method, which was trained based on full historical data that reflects how past orders actually behave. However, the MAPE for empirical is 74.14% indicating bad performing models, so we proceed using Machine Learning models instead.

For each model, we computed the RMSE, MAPE, and R-squared score, and evaluated both training and testing performance to ensure the selected model performs accurately and robustly. Based on the MAPE evaluation, we found that the best-performing model with the lowest MAPE is **Linear Regression**, which will be used in the Monte Carlo simulation.

Then we created a comparison table between Actual and Predicted Lead Time to display how far the forecast values are from the actual ones. The table indicates the errors between Actual vs Predict.

Model	Test_MAPE (%)
Linear Regression	54.46%
Random Forest Regressor	58.66%
XGBoost Regressor	54.85%

Overall Monte Carlo Simulation

We compared demand forecast accuracy across two horizons: 2 months ahead and 5 months ahead. A single SARIMA model was trained on monthly orders and combined with product choice, quantity, and lead-time predictions to generate Monte Carlo simulations. Actual demand for each horizon was used to compute MAPE on both order counts and total quantities.

Across both the end-month metric and average-window MAPE, the 2-month horizon consistently achieved lower error than the 5-month horizon, confirming that forecast accuracy deteriorates when predicting further into the future. This is consistent with our dataset, which exhibits highly inconsistent order patterns, including months with zero orders, making long-range forecasting inherently unreliable. For this business context, shorter horizons that rely on recent demand behavior provide significantly more stable and meaningful results. Therefore, reducing production lead time improves forecast reliability and strengthens inventory planning.

Horizon	Quantity Average MAPE	Order Count Average MAPE
2 Month	37.30%	72.73%

5 Month	1634.51%	5815.81%
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Business Recommendation

Given the highly irregular ordering patterns, frequent gaps in monthly demand, and the sharply deteriorating forecast accuracy observed for longer horizons, the company should **shorten its production and planning cycle from 5 months to 2 months or less** wherever operationally feasible. Near-term forecasting offers significantly higher reliability (2-month MAPE far lower than 5-month MAPE), making it better aligned with the volatility of this business.

In practical terms, this means prioritizing:

- **Smaller, more frequent production batches**
- **Shorter lead-time commitments with the supplier**
- **Rolling monthly demand reviews instead of long-range fixed plans**
- **Inventory policies driven by recent order behaviour rather than multi-month forecasts**

Shifting toward a shorter planning horizon reduces forecast error risk, avoids over-production or stockouts, and increases the company's ability to respond to rapidly changing demand patterns. In this environment, **operational flexibility is more valuable than long-range precision**, making short-term forecasting the most financially sound approach.